



DR. Amit Kumar Vishwakarma

AI Engineer / PhD (IIT Kanpur) / Ex-Ministry of Finance, India / Ex-Samsung R&D, Bangalore

CONTACT



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EDUCATION

Ph.D. in Electrical Engineering (Blockchain Technology)
Indian Institute of Technology Kanpur, India
(Joint Research with Tsinghua University, China)
2017 – 2022

Research Scholar (Visiting)
Tsinghua University, China
2019 – 2020

M.Tech. in Digital Systems
MNNIT Allahabad, India
2014 – 2016 | CGPA: 9.3/10

TECHNICAL SKILLS

Programming

- Python
- SQL
- JavaScript
- TypeScript
- HTML

AI & Machine Learning

- Machine Learning
- Deep Learning
- Generative AI
- LLMs
- AI Agents
- RAG
- Prompt Engineering

PROFILE SYNOPSIS

AI Engineer and Postdoctoral Researcher with expertise in Machine Learning, Generative AI, Large Language Models (LLMs), AI Agents, and intelligent decision support systems. Experienced in designing AI-driven solutions for supply chain optimization, sustainability, digital transformation, and public-sector applications through research and applied projects. Strong foundation in Python, predictive analytics, and blockchain-enabled systems, with a proven ability to translate advanced research into practical AI solutions. Passionate about building scalable, production-ready AI applications and continuously expanding expertise in modern AI engineering technologies.

WORK EXPERIENCE

Assistant Professor

Indian Institute of Management (IIM) Raipur, India | Mar 2026 – Present

- Teaching graduate courses in Data Analytics, AI, and Public Policy while developing hands-on AI applications, LLM-based solutions, and industry-focused projects in Generative AI and intelligent systems.
- Mentoring student projects in AI, Machine Learning, and data-driven business solutions.
- Conducting research on Generative AI, AI Agents, Blockchain, and sustainable digital transformation.

Postdoctoral Research Fellow – AI & Intelligent Systems

Khalifa University, Abu Dhabi, UAE | Mar 2024 – June 2025

- Developed AI-driven decision support systems for sustainability and supply chain.
- Designed machine learning models for carbon emissions prediction, optimization, and analytics.
- Conducted research on AI, blockchain, and digital twin technologies for enterprise applications.
- Collaborated with multidisciplinary teams on AI-enabled solutions for real-world challenges.

Assistant Professor (Artificial Intelligence & Machine Learning)

Thapar Institute of Engineering & Technology, India | Aug 2023 – Mar 2024

- Taught Artificial Intelligence, Machine Learning, and Blockchain Technology.
- Supervised AI and machine learning projects focused on predictive analytics and intelligent systems.
- Designed practical coursework integrating emerging AI technologies and industry use cases.

Technology Consultant (AI & Blockchain)

AJNIFM, Ministry of Finance, Government of India | Jul 2022 – Aug 2023

- Contributed to digital transformation initiatives using AI and blockchain technologies.
- Developed intelligent decision-support solutions for public finance and procurement.
- Worked with cross-functional stakeholders to evaluate technology adoption and automation strategies.

Research Associate

Samsung R&D Institute Bangalore, India | Jun 2021 – Dec 2021

- Conducted applied research on blockchain technologies and distributed systems.
- Supported the design and evaluation of secure, scalable software architectures.

AI Frameworks

- LangChain
- LangGraph
- Hugging Face Transformers
- PyTorch
- TensorFlow
- FastAPI

Databases

- PostgreSQL
- Redis
- ChromaDB
- FAISS
- Pinecone

Cloud & DevOps

- Docker
- Git
- Linux
- AWS
- Azure

Blockchain

- Ethereum
- Solidity
- Smart Contracts
- Hyperledger Fabric

LANGUAGES

English IELTS 6.5

Hindi (Native)

English

AWARDS

MHRD Scholarship

Chinese Government Scholarship



Projects

AI-Powered Multi-Regional Supply Chain Carbon Intelligence Platform

Tech Stack: Python, Machine Learning, Blockchain, Smart Contracts, Digital Twin

- Designed and developed an AI-powered decision support platform for monitoring and analysing carbon emissions across multi-regional supply chains.
- Developed machine learning models to forecast carbon emissions and identify opportunities for emission reduction and operational optimisation.
- Built a blockchain-enabled data layer to provide secure, transparent, and tamper-proof carbon emissions tracking.
- Implemented smart contracts to automate carbon credit allocation, compliance verification, and sustainability reporting.
- Delivered real-time analytics and decision support for ESG reporting and net-zero strategy planning.

Blockchain-Based AI Carbon Credit Trading Platform

Tech Stack: Python, Machine Learning, Blockchain, Ethereum, Solidity, Smart Contracts

- Designed and developed an AI- and blockchain-powered platform for transparent and auditable carbon credit trading.
- Developed AI models to analyse carbon emission trends and support data-driven carbon credit allocation and trading decisions.
- Implemented smart contracts to automate carbon credit issuance, validation, trading, and settlement processes.
- Built an immutable blockchain ledger to improve traceability, transparency, and regulatory compliance throughout the trading lifecycle.
- Enabled real-time analytics and monitoring of carbon credit transactions to support sustainability reporting and emissions management.

Blockchain-Based Peer-to-Peer Energy Trading Platform

Tech Stack: Python, Blockchain, Ethereum, Solidity, Smart Contracts

- Designed and developed a blockchain-enabled peer-to-peer energy trading platform for decentralised electricity markets.
- Implemented a credit-based payment mechanism using smart contracts to automate secure and trusted energy transactions.
- Developed a distributed ledger framework for transparent verification of energy generation, consumption, and trading records.
- Integrated virtual network architecture to support secure energy exchange between consumers and the power grid.

Certifications

- Advance AI Technique for the Supply Chain
- Fundamentals of Machine Learning for Supply Chain
- Introduction to Software Product Management
- Blockchain Technology and Cryptography Overview

Invited Talk

1. Tsinghua Energy Hub, China – Invited Speaker on AI and Blockchain for Peer-to-Peer Energy Trading.
2. Ministry of Finance, Government of India – Invited Speaker on Smart Contracts and Digital Transformation in Supply Chains.
3. Tsinghua International Blockchain Association (TIBA) – Invited Lecturer on Blockchain Applications, Research, and Emerging Challenges.

Publications

Journal Articles

- Applications of AI to Low-Carbon Decision Support Systems for Global Supply Chains, *Cleaner Logistics and Supply Chain*, 2025.
- Blockchain-Based Peer-to-Peer Renewable Energy Trading and Traceability of Transmission and Distribution Losses, *Journal of the Operational Research Society*, 2024.
- Developing a Delivery Prediction Model Using AI for Government e-Marketplace Contracts, *NIFM Journal of Public Financial Management*, 2025.
- Internet of Things Technology: Research and Challenges – A Survey, *Multimedia Tools and Applications*, 2024.